Bhaskar Neerati

Software Engineer | Python, FastAPI, PostgreSQL | github.com/bhaskarneerati
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SUMMARY

Software Engineer with 4+ years of experience building backend services and APIs in Python and FastAPI for banking and healthcare clients. Experience integrating REST APIs and conversational AI (Yellow.ai), building an event-driven invoice-processing platform with LLM agents (Lyzr) on AWS, and developing full-stack applications with Next.js. Comfortable debugging issues from logs and working from requirements to release in agile teams.

TECHNICAL SKILLS

Backend: Python, FastAPI, REST APIs, Pydantic, SQLAlchemy, Alembic, async programming (httpx, background tasks), Pytest

Databases: PostgreSQL, SQL, Redis, ChromaDB

AI / LLM: LangChain, Retrieval-Augmented Generation (RAG), embeddings, OpenAI / Gemini / Groq APIs, Lyzr agents, Yellow.ai, prompt engineering

Frontend: Next.js, React, TypeScript, JavaScript

Cloud and DevOps (working knowledge): AWS (Lambda, SQS, S3, ECS, RDS), Docker, GitHub Actions

Tools: Git, GitHub, Bitbucket, Claude Code (AI-assisted development)

EXPERIENCE

Phoebus AI Solutions | Senior Software Engineer Oct 2024 - Present

Invoice Processing Platform (client engagement for Lyzr)

Stack: Python, FastAPI, PostgreSQL, Next.js, AWS (ECS, Lambda, SQS, S3, RDS), Microsoft Graph API, Lyzr AI agents, SFTP

- Designed and built an event-driven platform that receives vendor invoices from Outlook mailboxes through Microsoft Graph change-notification webhooks (a Lambda renews subscriptions every 12 hours) and from a bulk-upload portal, to automate invoice entry into the client's Yardi accounting system.
- Implemented a queue-based pipeline (FastAPI API, SQS, separate worker) that extracts invoice fields with Lyzr AI agents, validates vendor, PO, property and duplicates against Yardi, and delivers import files over SFTP.
- Built a Next.js portal with a per-invoice step tracker, a "Needs Review" queue for exceptions, analytics, Microsoft SSO and email failure alerts.
- Deployed on AWS (ECS, RDS, S3, Amplify) with automated deployments from Bitbucket.

Healthcare Chatbot (Yellow.ai)

Stack: Yellow.ai, JavaScript, Python, REST APIs

- Built conversational flows, REST API integrations and custom JavaScript logic for a healthcare client's chatbot on Yellow.ai; delivered over about a year, serving 30,000+ users daily per the bot analytics, with the client reporting about 40% fewer manual support queries.
- Integrated 15+ REST APIs (appointments, insurance changes and more) into the conversation flows, with retry handling (up to 3 attempts) instead of failing on the first error, and logging of key request and response parameters to support debugging.
- Reduced average API response latency by about 30% (before and after averages from platform logs) by replacing a token-API call on every request with a global token and expiry check (Yellow App to Yellow Cloud migration), caching shared reference data (appointment types, specialties, clinic and provider lists, plan types) across users, and reusing policy details within a conversation to remove duplicate calls.
- Developed a Python script that automates RAG answer evaluation: it sends the client's test questions to the bot through Yellow.ai APIs and compares each answer with the expected one using similarity scoring, replacing about 2 days of manual checking per test round.

Accenture | Associate Software Engineer Sep 2022 - Oct 2024
Python Backend Developer, banking client (loan services) | Hyderabad (Hybrid)
Stack: Python, FastAPI, PostgreSQL, SQLAlchemy, Pydantic, Pytest, REST APIs, AWS Lambda, Docker, Jenkins, Git, Jira

- Developed and maintained FastAPI REST endpoints for a banking client's loan services, covering CRUD operations on loan data, business logic written from requirements, Pydantic request validation and consistent error handling.
- Worked with PostgreSQL through SQLAlchemy to read, insert and update application data; changed existing APIs to meet new requirements and integrated them with other backend services over REST.
- Wrote unit and API tests for the features I delivered; diagnosed and fixed defects reported by QA and found in testing by reading application logs and debugging Python code.
- Made small changes to existing Python AWS Lambda functions in the bank's event-driven transaction-processing flow (built on SQS and SNS), working with senior engineers.
- Supported production investigations and releases (including triggering the team's Jenkins pipelines) and worked from Jira tickets in a team of senior developers, a tech lead and QA engineers: daily stand-ups, sprint planning, code reviews.

PROJECTS

Prism: API Gateway & Developer Platform
Source: github.com/bhaskarneerati/project-prism | Live demo: project-prism-eight.vercel.app
Stack: Python, FastAPI, PostgreSQL, Redis, Next.js, React, Docker, GitHub Actions

- Built a self-hosted API gateway where developers register proxy routes and receive API keys; only SHA-256 hashes of the keys are stored.
- Implemented an async reverse proxy (FastAPI, httpx) that validates keys, enforces Redis sliding-window rate limits and monthly quotas, forwards requests, and records usage through non-blocking background tasks.
- Designed a PostgreSQL schema (6 tables, UUID keys, indexed lookups) managed with Alembic versioned migrations.
- Created a Next.js dashboard (Tailwind, shadcn/ui, Recharts) with usage analytics from aggregated async SQLAlchemy queries, request logs, and an admin panel.
- Deployed the backend on Render and the frontend on Vercel; GitHub Actions runs the Pytest suite on every push to main and blocks the deploy when tests fail.

RAG Assistant: Retrieval-Augmented Generation Chat Platform
Source: github.com/bhaskarneerati/rag-assistant
Stack: Python, FastAPI, LangChain, ChromaDB, sentence-transformers, OpenAI/Gemini/Groq, JavaScript

- Built a RAG system that lets users chat with private documents (PDF, Markdown, TXT), combining semantic vector search with LLM answers restricted to the retrieved context, with source citations.
- Designed a FastAPI backend with a modular ingestion pipeline (chunking, local embedding generation, repeat-safe indexing) feeding a persistent ChromaDB store.
- Added a provider-agnostic LLM layer (LangChain) supporting OpenAI, Gemini and Groq, selected through configuration.
- Built an admin dashboard for retrieval observability that shows, per session, the retrieved chunks and answers, backed by structured JSON logs across the RAG engine, vector DB and session layers.
- Created an embeddable chat widget in vanilla JS and CSS, loaded with a single script tag.

EDUCATION

Master of Science, Physics | University of Hyderabad 2020 - 2022
GPA: 8.54 / 10.0

Bachelor of Science (Hons.), Physics | Sri Sathya Sai Institute of Higher Learning 2017 - 2020
GPA: 7.7 / 10.0